# The Fractal Structure of Prime Numbers: Generation by Logical Operations

Harmonic transfer triangles, projective embedding, and the harbinger ledger

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**Abstract.** This paper studies the prefix probability vectors $P_k^{(r)} = (1/k)/H_r$ $(1 \le k \le r)$ induced by the harmonic numbers $H_r = \sum_{k=1}^{r} 1/k$, and builds on it an arithmetic dynamical system whose entire generation rule consists of two logical operations: compression $D$ and branching $B$. It is proved that the primes are exactly the "new-letter events" of this system: $r$ is prime if and only if the reduced denominator $\operatorname{den}(H_r)$ acquires a prime factor that has never appeared before, and that factor is $r$ itself (Theorem 7.1). The denominator anchors the primes ($p \mid \operatorname{den}(H_p)$, $p \nmid \operatorname{num}(H_p)$, $v_p(\operatorname{den}(H_p)) = 1$, and $p \mapsto H_p$ is injective on primes; Theorem 3.1), while the numerator carries Wolstenholme-type congruences ($p^2 \mid \operatorname{num}(H_{p-1})$ for $p \ge 5$; Theorem 3.2). The square-root embedding $z_k = \sqrt{P_k}$ sends the system to the unit sphere, where the Fisher metric becomes four times the spherical metric; the stepwise rotation angles satisfy the cosine telescope $\prod_{k<r} \cos\theta_k = 1/\sqrt{H_r}$, and the total budget $\Theta_r = \arccos(1/\sqrt{H_r})$ increases toward $\pi/2$ without ever reaching it (Theorems 6.2, 6.3). An interference form yields the period theorem: the superposition over the first $m$ primes has common period $2\pi \operatorname{lcm}(1, \ldots, p_m) = 2\pi e^{\psi(p_m)}$ (Theorem 6.4). The harbinger ledger $u_p := p^{-2} H_{p-1} \in \mathbb{Z}_p$ satisfies Glaisher's congruence $u_p \equiv -B_{p-3}/3 \pmod{p}$ (Theorem 7.2); its residue $r_p$ is empirically near-uniform over the 428 primes $5 \le p \le 2999$, while the denominator ledger connects rigorously to the prime number theorem and the Riemann hypothesis through $\psi(x)$ (the von Koch equivalence). It further proves an exact criterion for the erasure phenomenon (Proposition 7.4; exactly 40 complete and 15 partial erasures for $r \le 2000$), a precise formulation of the no-closed-form principle (the $\mathbb{Q}$-linear independence of $\{\ln p\}$ and the Mills-type tautology), and the computational boundary of the bare syntax: its symbolic dynamics has entropy rate zero, whereas the minimal extension allowing Fractran-style conditional duplication is Turing complete. Every numerical assertion has been independently verified with exact rational arithmetic; algorithms and data tables are collected in the appendices.

**Keywords:** harmonic numbers; prime distribution; transfer triangle; Fisher metric; Wolstenholme congruence; $p$-adic valuation; harbinger residue; erasure phenomenon; computational universality

## §1 Introduction

The harmonic numbers $H_r = \sum_{k=1}^{r} 1/k$ are among the oldest sequences in mathematics. Two of their properties have long been known. First, $H_r$ is never an integer for $r \ge 2$ (Theisinger's short proof of 1915, whose substance is a 2-adic valuation argument; [5]). Second, the set of prime factors of the reduced denominator $\operatorname{den}(H_r)$ evolves with $r$ in a highly irregular way: it stays unchanged most of the time, occasionally swallows a new factor, and very rarely drops an old one. The starting point of this paper is an exact characterization of the first phenomenon: the denominator absorbs a prime factor that has never occurred before if and only if $r$ is prime, and the absorbed factor is $r$ itself (Theorem 7.1). In other words, **the set of primes coincides with the set of alphabet-renewal events of the harmonic denominator sequence**. This equivalence rewrites the multiplicative question "what is the $r$-th prime" as an additive-valuative one: "does the $r$-th harmonic denominator change its alphabet".

Around this equivalence the paper does three things. First, it builds the generating system: the states are the prefix probability vectors $P_k^{(r)} = (1/k)/H_r$, and the only two operations are compression $D$ (multiplication by $\lambda_r = H_r/H_{r+1}$) and branching $B$ (adjoining $(1-\lambda_r)e_{r+1}$); §5 proves the generation rule $P^{(r+1)} = \lambda_r P^{(r)} \oplus (1-\lambda_r)e_{r+1}$ together with its perfect-memory, freezing, and orthogonality laws (§5). Second, it installs a geometric reading: the square-root embedding sends the states to the unit sphere, where the Fisher information metric becomes four times the spherical metric (§6; a standard fact of Amari–Nagaoka information geometry [12] in discrete coordinates); the cosines of the successive rotation angles telescope, $\prod_{k<r} \cos\theta_k = 1/\sqrt{H_r}$, and the total angular budget is permanently locked below $\pi/2$ (Theorems 6.2, 6.3); after attaching to each prime row a complex interference term $z_p(\varphi) = R_p e^{iR_p\varphi}$ with $R_p = 1 + H_p$, the superposition over the first $m$ primes has common period $2\pi\,\mathrm{lcm}(1,\ldots,p_m) = 2\pi e^{\psi(p_m)}$ (Theorem 6.4). Third, it installs a ledger reading: the denominator ledger meets the prime number theorem and the Riemann hypothesis through the Chebyshev function $\psi(x) = \ln\mathrm{lcm}(1,\ldots,\lfloor x\rfloor)$ (§8), while the numerator ledger is carried by the harbinger quantity $u_p = p^{-2}H_{p-1} \in \mathbb{Z}_p$, which satisfies Glaisher's 1900 congruence $u_p \equiv -B_{p-3}/3 \pmod p$ [2] and vanishes exactly at the Wolstenholme primes (§7).

The relation to the existing literature should be stated plainly. That $p^2$ divides the numerator of $H_{p-1}$ for $p \ge 5$ is Wolstenholme's 1862 theorem [1]; the congruence $H_{p-1} \equiv -\frac{p^2}{3}B_{p-3} \pmod{p^3}$ is Glaisher's 1900 result [2]; the equivalence of $\psi(x) \sim x$ with the prime number theorem, and of the Riemann hypothesis with $\psi(x) - x = O(\sqrt{x}\ln^2 x)$, are due to Chebyshev [3] and von Koch [4] respectively; the fact that the only Wolstenholme primes below $10^9$ are $16843$ and $2124679$ belongs to McIntosh and Roettger's 2007 computation [7]. These are all cited and give each a coherent place inside the system. The new contributions of the paper are: the organization of these classical facts into a single dynamical system generated by two logical operations; the new-letter equivalence (Theorem 7.1); the complete four-clause denominator anchoring with injectivity (Theorem 3.1); the cosine telescope and the angular budget (Theorems 6.2, 6.3); the period theorem (Theorem 6.4); the erasure criterion (Proposition 7.4); and the rigorous formulations of the no-closed-form principle and the computational boundary in §9–§10. Every assertion not currently proved — the equidistribution of harbinger residues, the limiting density of erasures, the composite-numerator problem — is explicitly marked as a conjecture or a numerical proposition, with its verification range stated.

All concrete numbers in the paper (Tables B.1–B.5) have been independently verified with exact rational arithmetic (arbitrary-precision fractions); congruence assertions are verified prime by prime in $\mathbb{Z}/p^k\mathbb{Z}$, with verification ranges stated in place, up to $p \le 2999$ (the 428 primes $p \ge 5$) and $r \le 2000$ (the erasure scan). Algorithms are given in Appendix A.

## §2 Preliminaries: harmonic numbers, prefix vectors, and alphabets

**Notation.** $H_0 := 0$ and $H_r := \sum_{k=1}^{r} 1/k$ $(r \ge 1)$; all fractions are assumed reduced, with $\mathrm{num}(x)$ and $\mathrm{den}(x)$ the numerator and denominator ($\mathrm{den}(x) > 0$); $v_q$ is the $q$-adic valuation ($v_q(0) = +\infty$); $\mathbb{Z}_p$ is the ring of $p$-adic integers; $B_n$ are the Bernoulli numbers ($B_2 = 1/6,\ B_4 = -1/30$); $\psi(x) := \sum_{p^a \le x} \ln p = \ln\mathrm{lcm}(1,\ldots,\lfloor x\rfloor)$ is the second Chebyshev function; $p$ always denotes a prime and $p_m$ the $m$-th prime.

**Definition 2.1 (prefix probability vector).** For $r \geq 1$ let

$$P^{(r)} := \frac{1}{H_r}\Big(1,\ \tfrac{1}{2},\ \tfrac{1}{3},\ \ldots,\ \tfrac{1}{r}\Big)\ \in\ \Delta^{r-1},$$

where $\Delta^{r-1} = \{x \in \mathbb{R}^r_{\geq 0} : \sum_k x_k = 1\}$ is the probability simplex. One calls coordinate $k$ the $k$-th **slot** and $P^{(r)}_k = (1/k)/H_r$ its **weight**.

**Definition 2.2 (alphabet and visible alphabet).** Let $L_r := \operatorname{lcm}(1, \ldots, r)$. The prime factor set of $L_r$ is $\mathcal{A}_r := \{p \text{ prime} : p \leq r\}$ (the full alphabet); the prime factor set of $\operatorname{den}(H_r)$ is denoted $\mathcal{V}_r$ (the **visible alphabet**). Clearly $\mathcal{V}_r \subseteq \mathcal{A}_r$, and $\mathcal{V}_1 = \emptyset$ (since $H_1 = 1$).

Section 3 establishes the anchoring of $\operatorname{den}(H_r)$ and $\operatorname{num}(H_r)$ at primes; §4 introduces the transfer triangle; §5 gives the operational semantics. The following elementary fact is used repeatedly, so it is isolated here.

**Lemma 2.3 (upper bound for the denominator valuation, and the equality condition).** For any prime $q$ and $r \geq 1$, write $a := \lfloor \log_q r \rfloor$ and $C := \lfloor r/q^a \rfloor$ (so $1 \leq C \leq q-1$). Then

$$v_q\big(\operatorname{den}(H_r)\big)\ \leq\ a,$$

with equality (i.e., $v_q(\operatorname{den}(H_r)) = a$) if and only if

$$S_C\ :=\ \sum_{c=1}^{C} \frac{1}{c} \not\equiv 0 \pmod q,$$

the sum being taken in $\mathbb{F}_q$ (the condition $c \leq q-1$ makes each term invertible). When $S_C \equiv 0 \pmod q$, the $q$-adic valuation of the denominator drops by the actual order of cancellation (the exact row-by-row formula is Proposition 7.4).

**Proof.** Write $H_r = N/L_r$ with $L_r = \operatorname{lcm}(1, \ldots, r)$ and $N = \sum_{k=1}^{r} L_r/k$. By the maximality of $a$ one has $v_q(L_r) = a$ and $q \nmid L_r/q^a$. Consider $N \bmod q$: if $v_q(k) \leq a-1$ then $q \mid L_r/k$ and the term vanishes; if $v_q(k) = a$ then $k = cq^a$ with $1 \leq c \leq C$ (by $k \leq r$ and the definition of $C$; $c \leq q-1$ forces $q \nmid c$), and $L_r/k = (L_r/q^a) \cdot c^{-1}$ is $q$-integral. Collecting terms,

$$N\ \equiv\ \frac{L_r}{q^a}\, S_C \pmod q, \qquad q \nmid \frac{L_r}{q^a}.$$

If $S_C \not\equiv 0 \pmod q$ then $q \nmid N$, reduction to lowest terms does not change the $q$-adic valuation of the denominator, and $v_q(\operatorname{den}(H_r)) = v_q(L_r) = a$; if $S_C \equiv 0 \pmod q$ then $q \mid N$ and cancellation lowers the denominator valuation by at least 1. The bound $v_q(\operatorname{den}(H_r)) \leq a$ holds in all cases because the reduced denominator always divides $L_r$. ∎

**Remark 2.4.** The case $S_C \equiv 0 \pmod q$ is exactly the erasure event studied in §7.3; its rarity (40 complete erasures for $r \leq 2000$, plus 15 partial ones) is explained by the criterion of Proposition 7.4. Two safe special cases will be used repeatedly: (i) when $r = q^a$ one has $C = 1$ and $S_1 = 1 \neq 0$, hence $v_q(\operatorname{den}(H_{q^a})) = a$; (ii) when $r = p = q$ is the same prime one has $a = 1$, $C = 1$, hence $v_p(\operatorname{den}(H_p)) = 1$. Erasure example: for $r = 6$, $q = 3$ one has $a = 1$, $C = 2$, $S_2 = 1 + 2^{-1} \equiv 1 + 2 = 0 \pmod 3$, and indeed $\operatorname{den}(H_6) = 20$ contains no factor 3.

## §3 Anchoring theorems: the denominator marks the primes, the numerator carries the congruences

This section proves the arithmetic foundation of the paper: at a prime row the reduced denominator $\mathrm{den}(H_r)$ is anchored by that prime with multiplicity exactly one, while the reduced numerator $\mathrm{num}(H_{p-1})$ is divisible by $p$ to Wolstenholme depth. The two properties point in opposite directions — the denominator marks where the primes are, the numerator records their congruence features — and this division of labor persists throughout the paper.

**Theorem 3.1 (denominator anchoring).** Let $p$ be prime. Then:
(i) $p \mid \mathrm{den}(H_p)$;
(ii) $p \nmid \mathrm{num}(H_p)$;
(iii) $p$ is the largest prime factor of $\mathrm{den}(H_p)$;
(iv) $v_p\big(\mathrm{den}(H_p)\big) = 1$;
(v) the map $p \mapsto H_p$ is injective on the set of primes.

**Proof.** Write $H_{p-1} = A/B$ in lowest terms. Since $\mathrm{den}(H_{p-1}) \mid L_{p-1}$ and $p \nmid L_{p-1}$, one has $p \nmid B$. Moreover

$$H_p = \frac{A}{B} + \frac{1}{p} = \frac{pA+B}{pB}, \qquad pA + B \equiv B \not\equiv 0 \pmod{p},$$

so the numerator is not divisible by $p$. Let $g = \gcd(pA+B, pB)$. Any power of $p$ dividing $g$ must divide $pA+B$, but $p \nmid pA+B$; hence $v_p(g) = 0$, and reduction to lowest terms changes neither $p$-adic valuation. Therefore $v_p(\mathrm{num}(H_p)) = v_p(pA+B) = 0$ (this is (ii)) and $v_p(\mathrm{den}(H_p)) = v_p(pB) = 1$ (this is (i) and (iv); note $v_p(B) = 0$). (iii): $\mathrm{den}(H_p) \mid L_p$, the prime factors of $L_p$ are exactly $\{q \text{ prime} : q \le p\}$, and by (i) the prime $p$ itself occurs in the denominator; hence $p$ is the largest prime factor. (v): if $p < q$ and $H_p = H_q$ then $0 = H_q - H_p = \sum_{p<k\le q} 1/k > 0$, a contradiction. ∎

**Remark 3.1.1.** Clause (iv) can also be read off from special case (ii) of Lemma 2.3 ($a = 1$, $C = 1$, $S_1 = 1 \not\equiv 0$). The two routes agree: $p^2 > p$ guarantees that row $p$ introduces only the first power of $p$.

**Theorem 3.2 (numerator anchoring; Wolstenholme).** Write $H_{p-1} = A/B$ in lowest terms. Then:
(i) for $p \ge 3$, $p \mid A$;
(ii) (Wolstenholme [1]) for $p \ge 5$, $p^2 \mid A$; equivalently $v_p(H_{p-1}) \ge 2$.

**Proof.** Work in $\mathbb{Z}_p$ (every term of $H_{p-1}$ has denominator coprime to $p$, so $H_{p-1} \in \mathbb{Z}_p$, and $v_p(H_{p-1}) = v_p(A) - v_p(B) = v_p(A)$). Put $m = (p-1)/2$. Mirror pairing $j \leftrightarrow p - j$ gives

$$H_{p-1} = \sum_{j=1}^{m} \left(\frac{1}{j} + \frac{1}{p-j}\right) = p \sum_{j=1}^{m} \frac{1}{j(p-j)} =: pT, \qquad T \in \mathbb{Z}_p,$$

which already proves (i): $v_p(H_{p-1}) \ge 1$. For (ii), reduce $T$ mod $p$ in $\mathbb{Z}_p$: since $1/(p-j) \equiv -1/j \pmod{p}$,

$$T \;\equiv\; -\sum_{j=1}^{m} \frac{1}{j^2} \;=\; -H_m^{(2)} \pmod{p}, \qquad H_m^{(s)} := \sum_{j=1}^{m} j^{-s}.$$

Pair once more: $1/(p-j)^2 \equiv 1/j^2 \pmod{p}$, hence $2H_m^{(2)} \equiv H_{p-1}^{(2)} \pmod{p}$. The map $j \mapsto j^{-1} \bmod p$ permutes

$(\mathbb{Z}/p\mathbb{Z})^\times$, so

$$H_{p-1}^{(2)} \equiv \sum_{a=1}^{p-1} a^2 = \frac{(p-1)p(2p-1)}{6} \equiv 0 \pmod p,$$

the last step because $p \nmid 6$ for $p \geq 5$ while the numerator contains the factor $p$. Hence $T \equiv 0 \pmod p$ and $v_p(H_{p-1}) \geq 2$. ∎

**Remark 3.2.1 (Wolstenholme primes).** Primes with $v_p(H_{p-1}) \geq 3$ (equivalently $\binom{2p-1}{p-1} \equiv 1 \pmod{p^4}$) are called Wolstenholme primes. Only two are known below $10^9$, namely 16843 and 2124679 [7]; whether there are infinitely many is open (§11). In the ledger language of this paper they are the zeros of the harbinger residue $r_p$ (Definition 7.2).

The composite direction is opposite to the prime direction and forms the dual picture of Theorem 3.2. It cannot be proved in full, and is stated as a numerical proposition with its verification range made explicit.

**Proposition 3.3 (composite numerators; numerical proposition).** For every composite $n$ with $4 \leq n \leq 2000$, $n \nmid \operatorname{num}(H_{n-1})$. Verification: $H_{n-1}$ was computed and reduced in exact rational arithmetic for each $n$ (Appendix A, Algorithm A.2); the number of mismatches is zero.

**Conjecture 3.4.** For every composite $n \geq 4$, $n \nmid \operatorname{num}(H_{n-1})$. Combined with Theorem 3.2(i) (odd primes always satisfy the divisibility) and the trivial exception $n = 2$ ($H_1 = 1$), this is equivalent to: for $n \geq 2$, $n \mid \operatorname{num}(H_{n-1})$ if and only if $n$ is an odd prime — the divisibility is a primality property. The conjecture is of the same kind as Eswarathasan and Levine's study [6] of $p$-integral harmonic sums; in a related direction, the converse of the binomial form of Wolstenholme's theorem (whether a composite $n$ can satisfy $\binom{2n-1}{n-1} \equiv 1 \pmod{n^3}$) was posed and studied by McIntosh [8], but the present first-order form of the statement is, to the authors' knowledge, not made explicitly in the literature.

## §4 The transfer triangle: moment law, difference law, and the $p$-adic entrance law

**Definition 4.1 (transfer triangle).** The infinite lower-triangular array

$$T(r,k) := \frac{1}{k}, \qquad 1 \leq k \leq r,$$

whose $r$-th row is the unnormalized form of the state $P^{(r)}$; the row sum is $\sum_{k=1}^{r} T(r,k) = H_r$.

The triangle has three structural laws, belonging respectively to analysis, algebra, and arithmetic.

**Proposition 4.2 (moment law and complete multiplicativity).** (i) $T(r,k) = \displaystyle\int_0^1 x^{k-1}\,\mathrm{d}x$: column $k$ is the moment of $x^{k-1}$ under Lebesgue measure; (ii) $H_r = \displaystyle\int_0^1 \frac{1-x^r}{1-x}\,\mathrm{d}x$; (iii) the sequence $a_k = 1/k$ is completely multiplicative: $a_{mn} = a_m a_n$ for all $m, n \geq 1$.

**Proof.** (i) $\int_0^1 x^{k-1}\mathrm{d}x = [x^k/k]_0^1 = 1/k$. (ii) Integrate the finite geometric series $\sum_{k=0}^{r-1} x^k = (1-x^r)/(1-x)$ term by term and apply (i). (iii) $a_m a_n = 1/(mn) = a_{mn}$. ∎

**Remark 4.2.1.** Clause (iii) is what distinguishes the harmonic triangle from the Pascal triangle ($a_k = 1$, additive) and the Stirling arrays: complete multiplicativity gives the prime slots the status of generators — the values $\{1/p\}$ at primes generate all values by multiplication. This is the algebraic shadow of "prime = new letter" (§7).

**Proposition 4.3 (Leibniz difference law).** (i) $\dfrac{1}{k} = \dfrac{1}{k+1} + \dfrac{1}{k(k+1)}$; (ii) (telescoping) $\dfrac{1}{r} = \displaystyle\sum_{k=r}^{\infty} \dfrac{1}{k(k+1)}$, the series converging in the usual sense.

**Proof.** (i) Common denominator: $(k+1)/(k(k+1)) = 1/k$. (ii) By (i), the partial sum $\sum_{k=r}^{N} 1/(k(k+1)) = \sum_{k=r}^{N}(1/k - 1/(k+1)) = 1/r - 1/(N+1) \to 1/r$. ∎

**Remark 4.3.1.** Proposition 4.3 is the difference structure of Leibniz's harmonic triangle from the 1679 *Characteristica geometrica* [14]: each cell is the sum of the cell directly below and the cell below-right. In this language, (i) reads "the weight of slot $k$ = the part it cedes to the next slot + an indivisible remainder"; this is the discrete prototype of the compression/branching operations of §5.

**Proposition 4.4 (the $p$-adic entrance law).** For every prime $q$, $v_q(T(r,k)) = -v_q(k)$, independently of the row index $r$; in particular, the prime $q$ first appears as a denominator factor of the triangle at the cell $(r,k) = (q,q)$, and column $k$ carries exactly $q^{-a}$ on the rows with $q^a \mid k$.

**Proof.** The $q$-adic valuation of $T(r,k) = 1/k$ is determined by $k$ alone: $v_q(1/k) = -v_q(k)$. The smallest $k$ with $q \mid k$ is $q$ itself, and the cell $(r,k)$ exists only for $r \geq k$; hence the earliest entrance cell is $(q,q)$. ∎

Proposition 4.4 states that the arithmetic of a single cell is trivial; all structure emerges in the reduction of the row sums $H_r$ — whether the columns' $q^{-a}$ cancel or survive the summation is adjudicated by the equality condition of Lemma 2.3, and erasure (§7.3) is precisely the event "survival fails".

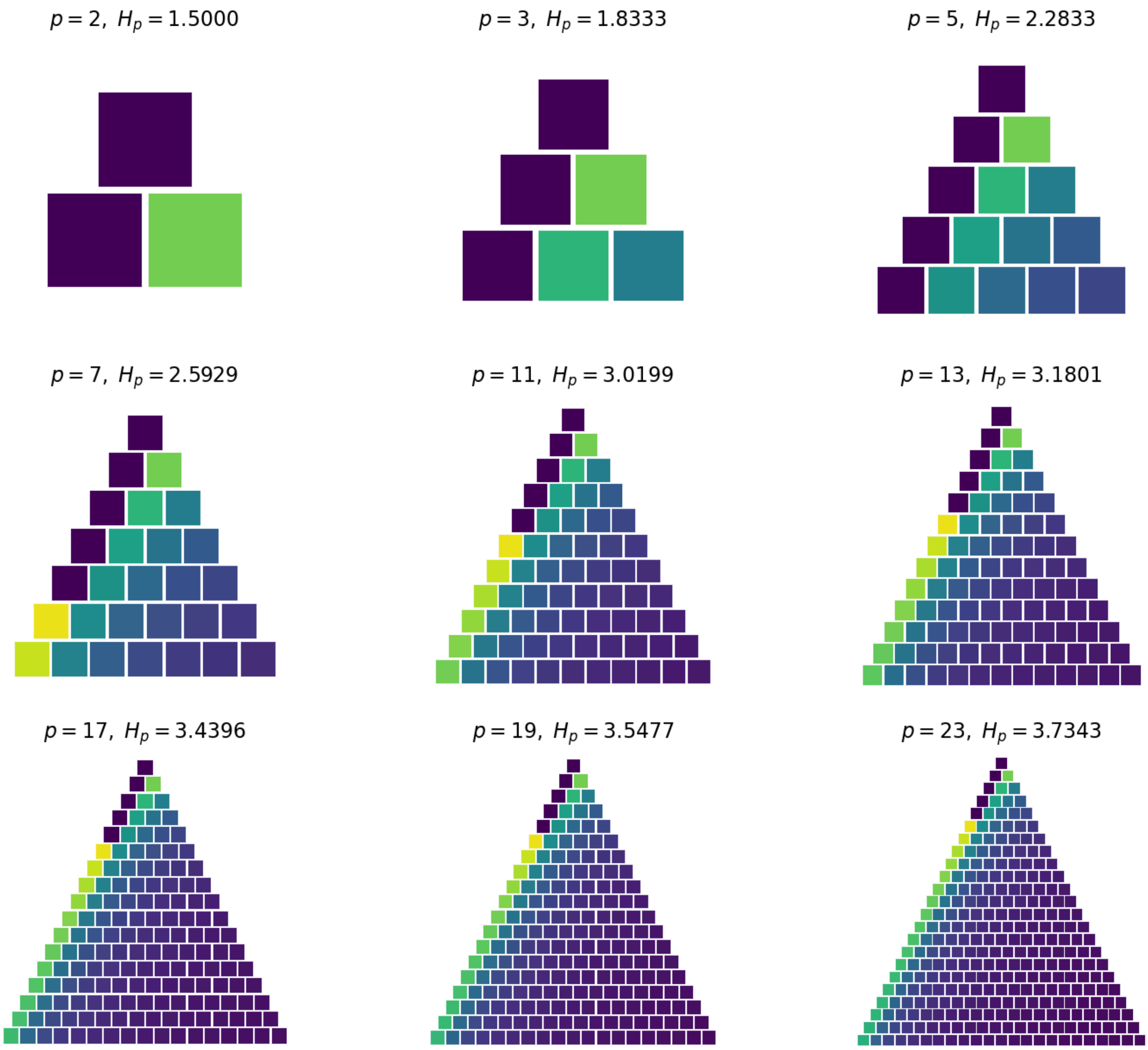


**Figure 1.** Transfer triangles of the first nine prime rows ($p = 2, 3, 5, \ldots, 23$). Cell $(r, k)$ has height $1/k$ and is colored by the normalized weight $P_k^{(r)} = (1/k)/H_r$. A new prime $p$ enters along the diagonal cell $(p, p)$ (Proposition 4.4); its weight $1/(pH_p)$ freezes rapidly as $p$ grows (Proposition 5.4).

## §5 Operational semantics: compression $D$, branching $B$, and the generation rule

This section decomposes the passage from $P^{(r)}$ to $P^{(r+1)}$ into two elementary operations and proves the three laws governing the recursion. Write $\lambda_r := H_r/H_{r+1}$ ($r \geq 1$), and identify $\Delta^{r-1}$ with its image in $\Delta^r$ under $x \mapsto (x, 0)$.

**Definition 5.1 (compression and branching).** For $x \in \Delta^{r-1}$ define

$$D_r(x) := \lambda_r\, x \;\in\; \mathbb{R}^r_{\geq 0}, \qquad B_r(x) := D_r(x) \oplus \bigl(1 - \lambda_r\bigr) e_{r+1} \;\in\; \Delta^r,$$

where $e_{r+1} = (0, \ldots, 0, 1)$. One calls $D_r$ **compression** (it multiplies all old weights by $\lambda_r$) and $B_r$ **branching** (it concentrates the ceded total mass $1 - \lambda_r$ on the new slot $r + 1$).

**Proposition 5.2 (generation rule).** For every $r \geq 1$,

$$P^{(r+1)} \;=\; \lambda_r\, P^{(r)} \;\oplus\; \big(1-\lambda_r\big)\, e_{r+1} \;=\; B_r\big(P^{(r)}\big),$$

and $1-\lambda_r = \dfrac{1/(r+1)}{H_{r+1}}$ is exactly the normalized weight of the new slot.

**Proof.** For $k \leq r$: $\lambda_r P_k^{(r)} = \dfrac{H_r}{H_{r+1}} \cdot \dfrac{1/k}{H_r} = \dfrac{1/k}{H_{r+1}} = P_k^{(r+1)}$. For $k = r+1$: $1-\lambda_r = \dfrac{H_{r+1}-H_r}{H_{r+1}} = \dfrac{1/(r+1)}{H_{r+1}} = P_{r+1}^{(r+1)}$. The two sides agree coordinatewise, and both sum to 1. ∎

**Proposition 5.3 (perfect-memory law).** For every $r \geq 1$ and $k \leq r$,

$$\frac{P_k^{(r+1)}}{P_k^{(r)}} \;=\; \lambda_r,$$

the ratio being independent of $k$; equivalently, restricting $P^{(r+1)}$ to its first $r$ slots and renormalizing recovers $P^{(r)}$ exactly. Hence the entire history $\{P^{(j)}\}_{j\leq r}$ can be inverted from $P^{(r)}$ without distortion, layer by layer.

**Proof.** A direct corollary of Proposition 5.2: $P_k^{(r+1)} = \lambda_r P_k^{(r)}$ for $k \leq r$, and the ratio $\lambda_r$ does not involve $k$. Inversion: $P_k^{(r)} = P_k^{(r+1)}\big/\big(1-P_{r+1}^{(r+1)}\big)$, since $\lambda_r = 1 - P_{r+1}^{(r+1)}$. ∎

**Proposition 5.4 (freezing law).** $\lambda_r = 1 - \dfrac{1}{(r+1)H_{r+1}}$ is strictly increasing in $r$, and $\lambda_r \uparrow 1$ as $r \to \infty$.

**Proof.** $(r+1)H_{r+1} - rH_r = r/(r+1) + H_r > 0$, so $t_r := (r+1)H_{r+1}$ is strictly increasing, $1/t_r$ is strictly decreasing, and $\lambda_r = 1 - 1/t_r$ is strictly increasing. Since $t_r \to \infty$ (as $H_r \to \infty$), the limit is 1. ∎

**Proposition 5.5 (orthogonality law).** In the standard inner product of $\mathbb{R}^{r+1}$, $D_r\big(P^{(r)}\big) \perp e_{r+1}$, and consequently

$$\big\|P^{(r+1)}\big\|^2 \;=\; \lambda_r^2\, \big\|P^{(r)}\big\|^2 + \big(1-\lambda_r\big)^2 :$$

each step splits the mass at a right angle.

**Proof.** The $(r+1)$-st coordinate of $D_r(P^{(r)})$ is 0, so it is orthogonal to $e_{r+1}$; the norm identity is the Pythagorean theorem. ∎

**Remark 5.5.1.** The three laws divide the labor as follows: the perfect-memory law guarantees losslessness (reversibility), the freezing law guarantees asymptotic stability of the old slots ($\lambda_r \to 1$), and the orthogonality law guarantees geometric independence of old and new mass. Section 6 will show that orthogonal splitting becomes rotation through a constant angle under the square-root embedding.

## §6 The square-root embedding: Fisher metric, cosine telescope, and the period theorem

### 6.1 Embedding and metric

**Definition 6.1 (square-root embedding).** $z^{(r)} := \big(\sqrt{P_1^{(r)}}, \ldots, \sqrt{P_r^{(r)}}\big) \in S^{r-1}$ (the unit sphere in the first orthant).

**Proposition 6.1.1 (pullback of the Fisher metric).** The pullback of the Fisher–Rao metric $g = \sum_k \dfrac{\mathrm{d}P_k \otimes \mathrm{d}P_k}{P_k}$ on $\Delta^{r-1}$ under the map $P = z^2$ (coordinatewise squaring) is

$$g\big|_z \;=\; 4\sum_{k=1}^{r} \mathrm{d}z_k \otimes \mathrm{d}z_k,$$

i.e., four times the standard spherical metric. The square-root embedding is therefore a local isometry up to the constant factor $2$.

**Proof.** $P_k = z_k^2$ gives $\mathrm{d}P_k = 2z_k\,\mathrm{d}z_k$; substituting, $\mathrm{d}P_k^2/P_k = 4z_k^2\mathrm{d}z_k^2/z_k^2 = 4\,\mathrm{d}z_k^2$. ∎

**Remark 6.1.2.** This is a standard fact of information geometry (Amari & Nagaoka [12], §2): in square-root coordinates the probability simplex becomes a sphere and the Fisher metric becomes Euclidean. What is needed is its consequence: the D/B recursion acquires an angular interpretation on the sphere.

### 6.2 Rotation angles, the telescope, and the total budget

**Theorem 6.2 (stepwise angles and the cosine telescope).** Let $\theta_r \in (0, \pi/2)$ be the spherical angle between $z^{(r)}$ and $z^{(r+1)}$ (viewed in $\mathbb{R}^{r+1}$, with $z^{(r)}$ padded by a zero coordinate), and let $\Theta_r \in (0, \pi/2)$ be the angle between $z^{(1)} = e_1$ and $z^{(r)}$. Then for every $r \geq 1$:

$$\cos\theta_r = \sqrt{\lambda_r}, \qquad \tan\theta_r = \frac{1}{\sqrt{(r+1)\,H_r}},$$

$$\cos\Theta_r \;=\; \prod_{k=1}^{r-1}\cos\theta_k \;=\; \frac{1}{\sqrt{H_r}}.$$

**Proof.** Inner product: $\langle z^{(r+1)}, z^{(r)} \oplus 0\rangle = \sum_{k=1}^r \sqrt{P_k^{(r+1)}P_k^{(r)}} = \sqrt{\lambda_r}\sum_{k=1}^r P_k^{(r)} = \sqrt{\lambda_r}$ (Proposition 5.3); both vectors are unit, so $\cos\theta_r = \sqrt{\lambda_r}$. Tangent:

$$\tan^2\theta_r = \frac{1-\lambda_r}{\lambda_r} = \frac{(1/(r+1))/H_{r+1}}{H_r/H_{r+1}} = \frac{1}{(r+1)H_r}.$$

Telescope: $\prod_{k=1}^{r-1}\cos^2\theta_k = \prod_{k=1}^{r-1}\lambda_k = \prod_{k=1}^{r-1}\dfrac{H_k}{H_{k+1}} = \dfrac{H_1}{H_r} = \dfrac{1}{H_r}$. On the other hand $\langle e_1, z^{(r)}\rangle = z_1^{(r)} = \sqrt{P_1^{(r)}} = 1/\sqrt{H_r}$. The two expressions agree, and $\cos\Theta_r := \langle e_1, z^{(r)}\rangle$, so all three quantities coincide. ∎

**Theorem 6.3 (locking of the total budget).** $\Theta_r = \arccos(1/\sqrt{H_r})$ is strictly increasing in $r$; $\sup_r \Theta_r = \pi/2$, and $\Theta_r < \pi/2$ for every finite $r$.

**Proof.** $H_r$ strictly increasing $\Rightarrow 1/\sqrt{H_r}$ strictly decreasing $\Rightarrow$ arccos strictly increasing. $H_r \to \infty$ gives $\sup = \arccos(0) = \pi/2$; finiteness of $H_r$ at finite $r$ gives the strict inequality. ∎

**Remark 6.3.1.** The bound $\pi/2$ carries no arbitrariness of constant choices: it is the spherical translation of "the first slot's weight $1/H_r \to 0$". The initial direction $e_1$ is the only memory of the system that does not die out; the orbit becomes asymptotically orthogonal to it without ever arriving. Figure 2 shows the approach curve of $\Theta_r$.

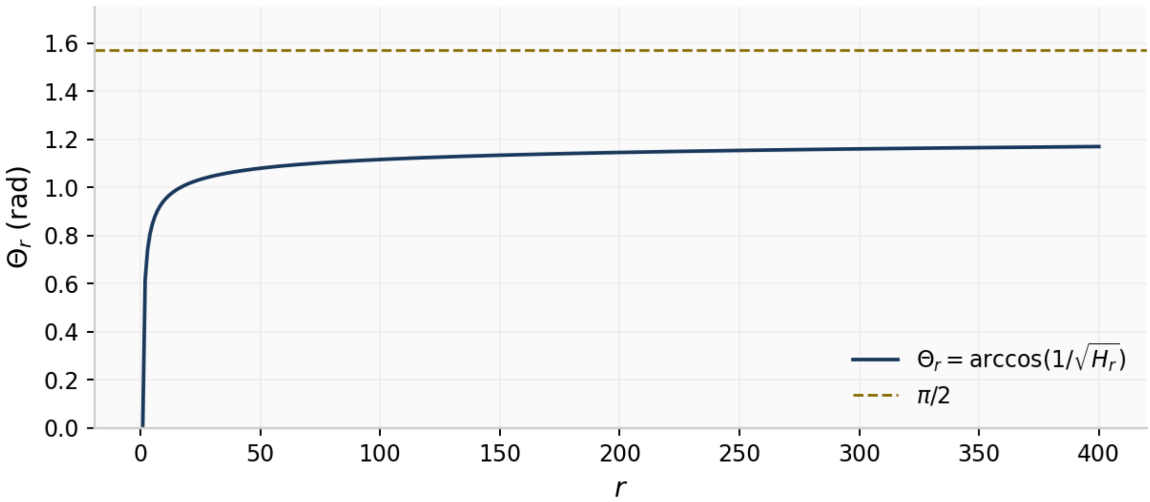


**Figure 2.** The total angular budget $\Theta_r = \arccos(1/\sqrt{H_r})$ $(1 \le r \le 400)$ approaching $\pi/2$ (dashed) monotonically; by Theorem 6.3 the approach is strict and never attained.

### 6.3 The interference form and the period theorem

**Definition 6.3.1 (interference term of a prime row).** For each prime $p$ define the complex-valued function

$$z_p(\varphi) \;:=\; R_p\, e^{iR_p\varphi}, \qquad R_p := 1 + H_p,$$

and the superposed intensity over the first $m$ primes

$$I_m(\varphi) \;:=\; \Big|\sum_{j=1}^{m} z_{p_j}(\varphi)\Big|^2.$$

The reading of $R_p = 1 + H_p$: $H_p$ is the total weight of row $p$ (including the new slot), and the $+1$ is the one new unit that the row contributes in the sense of the alphabet (the new letter $p$ itself, Theorem 7.1); amplitude and frequency are both set to $R_p$, so that each prime contributes one oscillator in which mass and frequency coincide.

**Theorem 6.4 (period theorem).** Let $m \geq 3$. $I_m$ is a trigonometric polynomial in $\varphi$, and its minimal positive period is

$$T_m \;=\; 2\pi\ \mathrm{lcm}(1, 2, \ldots, p_m) \;=\; 2\pi\, e^{\psi(p_m)}.$$

In particular, since the prime number theorem gives $\psi(p_m) \sim p_m$, the period grows like $e^{(1+o(1))p_m}$. (The cases $m \leq 2$ are treated in Remark 6.4.1.)

**Proof.** Step 1 (frequency decomposition and the minimal-period formula). Expand $I_m = \sum_{j,l} R_{p_j} R_{p_l} e^{i(R_{p_j} - R_{p_l})\varphi}$. Since $R_p = 1 + H_p$, the frequency set is the difference set $\{R_{p_j} - R_{p_l}\} = \{H_{p_j} - H_{p_l}\} \subset \mathbb{Q}$. All terms sharing one frequency $\omega$ have positive real coefficients $R_{p_j} R_{p_l}$, so no cancellation occurs and every attained difference frequency is genuinely present in $I_m$. By the linear independence of the exponentials, $T$ is a period of $I_m$ if and only if $T\omega \in 2\pi\mathbb{Z}$ for every frequency present. Write $T = 2\pi\tau$: the diagonal terms $j = l$ give zero frequency and impose no constraint, and a reduced nonzero frequency $\omega$ forces $\mathrm{den}(\omega) \mid \tau$. Hence the minimal positive period is $T_m = 2\pi\tau_m$, where

$$\tau_m \;=\; \mathrm{lcm}_{1 \leq j,l \leq m}\, \mathrm{den}\left(H_{p_j} - H_{p_l}\right).$$

Step 2 ($\tau_m = \mathrm{lcm}(1, \ldots, p_m)$). Write $L$ for the right side. "$\tau_m \mid L$": $\mathrm{den}(H_{p_j} - H_{p_l}) \mid \mathrm{lcm}(\mathrm{den}(H_{p_j}), \mathrm{den}(H_{p_l})) \mid L$. " $L \mid \tau_m$": take any prime power $q^a \mid L$ (i.e., $q^a \leq p_m$); it suffices to exhibit a pair $(r, s)$ inside $\{p_1, \ldots, p_m\}$ with $v_q(\mathrm{den}(H_r - H_s)) \geq a$. A valuation comparison fact is used: if $A, B \in \mathbb{Q}$ satisfy $v_q(\mathrm{den}(A)) = \alpha > \beta = v_q(\mathrm{den}(B))$, then $q \nmid \mathrm{num}(A)$ by reducedness, and in the common-denominator form of $A - B$ the numerator has $q$-adic valuation $\beta$ (the two terms have valuations $\beta$ and $\geq \alpha$, the minimum uniquely attained), so $v_q(\mathrm{den}(A - B)) = \alpha$ after reduction. Four cases. (i) $q$ odd, $a = 1$: take $(r, s) = (q, 2)$ — special case (ii) of Lemma 2.3 gives $v_q(\mathrm{den}(H_q)) = 1$, while $v_q(\mathrm{den}(H_2)) = 0$, and the comparison fact yields $v_q(\mathrm{den}(H_q - H_2)) = 1$. (ii) $q$ odd, $a \geq 2$: let $p'$ be the smallest prime $\geq q^a$; by the Bertrand–Chebyshev theorem (for $x \geq 1$ there is always a prime in $(x, 2x]$ [3]; see also [13], Theorem 418) $q^a < p' < 2q^a$, and $p' \leq p_m$ ($p_m$ itself is a candidate), so $p' = p_{j'}$ for some $j' \leq m$. Take $(r, s) = (p', q)$: special case (i) of Lemma 2.3 ($\lfloor p'/q^a \rfloor = 1$, $C = 1$, $S_1 = 1$) gives $v_q(\mathrm{den}(H_{p'})) = a$, while $v_q(\mathrm{den}(H_q)) = 1 < a$ (the only power of $q$ not exceeding $q$ is $q$ itself, and the same special case applies), so the comparison fact gives $v_q(\mathrm{den}(H_{p'} - H_q)) = a$. (iii) $q = 2$, $a = 1$: take $(r, s) = (5, 2)$ ($m \geq 3$ puts 5 in the index set): $H_5 - H_2 = 47/60$ with $v_2(60) = 2 \geq 1$. (iv) $q = 2$, $a \geq 2$: let $p'$ be the smallest prime $\geq 2^a$ ($2^a < p' < 2^{a+1}$, $p' \leq p_m$), paired with $s = 3$: $v_2(\mathrm{den}(H_{p'})) = a$ (special case (i) of Lemma 2.3) and $v_2(\mathrm{den}(H_3)) = 1 < a$, so $v_2(\mathrm{den}(H_{p'} - H_3)) = a$ by the comparison fact. The indices $2, 3, 5, q, p'$ used in the four cases are all $\leq p_m$ (this is where $m \geq 3$ is used). Altogether $q^a \mid \tau_m$ for every $q^a \parallel L$, hence $L \mid \tau_m$. Step 3 (translation into $\psi$). $\mathrm{lcm}(1, \ldots, n) = \prod_{q \leq n} q^{\lfloor \log_q n \rfloor} = e^{\psi(n)}$ is the defining identity of $\psi$. The growth rate is the equivalent form $\psi(x) \sim x$ of the prime number theorem (founded in [3]; a standard proof is in [13], Chapter XXII). ∎

**Remark 6.4.1 (the cases $m \leq 2$).** For $m = 1$, $I_1(\varphi) \equiv R_2^2$ is constant and has no minimal positive period. For $m = 2$ the only nonzero difference frequency is $H_3 - H_2 = -1/3$, hence $T_2 = 6\pi$; the value $12\pi$ corresponding to $\mathrm{lcm}(\mathrm{den}(H_2), \mathrm{den}(H_3)) = 6$ is the minimal period of the superposition $z_2 + z_3$ itself, not of $I_2 = |z_2 + z_3|^2$ — $I_m$ sees only difference frequencies, which is exactly why $m = 2$ behaves differently from $m \geq 3$. The difference-frequency decomposition of Step 1 is valid for every $m$; the hypothesis $m \geq 3$ is used only in case (iii) of Step 2.

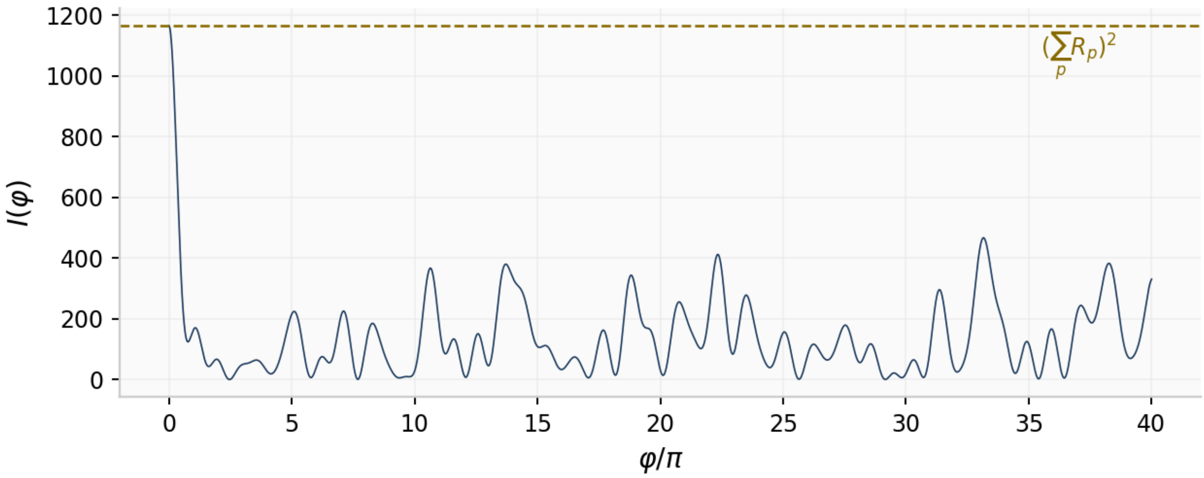


**Figure 3.** Interference intensity $I_9(\varphi) = |\sum_{j\le 9} R_{p_j} e^{iR_{p_j}\varphi}|^2$ of the first nine primes ($R_p = 1 + H_p$); the horizontal axis is $\varphi/\pi$. The dashed line marks the coherent maximum $(\sum R_{p_j})^2 = 1164.93$; by Theorem 6.4 the pattern repeats with exact period $2\pi\,\mathrm{lcm}(1,\dots,23) = 2\pi \cdot 5354228880$.

## §7 Prime events and the harbinger ledger

### 7.1 The new-letter equivalence

Recall that $\mathcal{V}_r$ is the prime factor set of $\mathrm{den}(H_r)$. It is said that a **new-letter event** occurs at $r$ if $\mathcal{V}_r$ contains a prime factor that no earlier row has ever contained: $\mathcal{V}_r \not\subseteq \bigcup_{j<r} \mathcal{V}_j$.

**Theorem 7.1 (new letters $\Longleftrightarrow$ primes).** (i) $\bigcup_{j\le r} \mathcal{V}_j = \mathcal{A}_r = \{q \text{ prime} : q \le r\}$; (ii) a new-letter event occurs at $r$ if and only if $r$ is prime; the newly appeared prime factor is then unique, and it is $r$ itself: $\mathcal{V}_r \setminus \bigcup_{j<r} \mathcal{V}_j = \{r\}$.

**Proof.** (i) "$\subseteq$": $\mathcal{V}_j \subseteq \mathcal{A}_j \subseteq \mathcal{A}_r$. "$\supseteq$": for a prime $q \le r$, Theorem 3.1(i) gives $q \mid \mathrm{den}(H_q)$, i.e., $q \in \mathcal{V}_q \subseteq \bigcup_{j\le r} \mathcal{V}_j$. (ii) If $r = p$ is prime: $p \in \mathcal{V}_p$ (Theorem 3.1(i)), while every $j < p$ satisfies $\mathcal{V}_j \subseteq \mathcal{A}_j$, whose elements are $\le j < p$; hence $p \notin \bigcup_{j<p} \mathcal{V}_j$. Uniqueness: any other element $q$ of $\mathcal{V}_p$ satisfies $q < p$ (Theorem 3.1(iii)), and by (i) it appeared already in $\mathcal{V}_q$. If $r$ is composite: $\mathcal{V}_r \subseteq \mathcal{A}_r = \mathcal{A}_{r-1}$ ($r$ itself is not prime, so $\mathcal{A}_r$ and $\mathcal{A}_{r-1}$ coincide), and by (i), $\mathcal{A}_{r-1} = \bigcup_{j\le r-1} \mathcal{V}_j$; hence no new letter. ∎

**Remark 7.1.1.** The theorem characterizes first appearances, not row-by-row differences: $\mathcal{V}_r \setminus \mathcal{V}_{r-1}$ may contain the re-infiltration of an old letter (example: $\mathcal{V}_6 = \{2, 5\}$, $\mathcal{V}_7 = \{2, 3, 5, 7\}$, and the $3$ in the difference $\{3, 7\}$ is a re-infiltration; see §7.3). Formulating the criterion as "difference against the entire history" is essential — this is exactly where the equivalence is non-trivial.

## 7.2 The harbinger congruence

**Definition 7.2 (harbinger quantity and harbinger residue).** For $p \geq 5$, Theorem 3.2(ii) gives $u_p := p^{-2} H_{p-1} \in \mathbb{Z}_p$; one calls $u_p$ the harbinger quantity and its residue class

$$r_p \ := \ u_p \bmod p \ \in \ \{0, 1, \ldots, p-1\}$$

(represented by the least non-negative residue) the harbinger residue.

**Theorem 7.2 (the harbinger congruence; Glaisher [2]).** For $p \geq 5$,

$$H_{p-1} \ \equiv \ -\frac{p^2}{3} B_{p-3} \pmod{p^3}, \qquad \text{i.e.,} \qquad u_p \ \equiv \ -\frac{B_{p-3}}{3} \pmod{p}.$$

In particular, $r_p = 0$ if and only if $p$ is a Wolstenholme prime (Remark 3.2.1).

**Proof.** Step 1 (exact reduction inside the system). Put $m = (p-1)/2$. From $1/(p-j) = -j^{-1} \sum_{t \geq 0} (p/j)^t$ (convergent in $\mathbb{Z}_p$),

$$H_{p-1} = p \sum_{j=1}^{m} \frac{1}{j(p-j)} = -\sum_{t \geq 0} p^{t+1} H_m^{(t+2)} \ \equiv \ -pH_m^{(2)} - p^2 H_m^{(3)} \pmod{p^3},$$

where $H_m^{(s)} := \sum_{j=1}^{m} j^{-s} \in \mathbb{Z}_p$ and the terms with $t \geq 2$ carry the factor $p^3$. On the other hand, $(p-j)^{-2} = j^{-2}(1 - p/j)^{-2} \equiv j^{-2} + 2p\, j^{-3} \pmod{p^2}$, and summation gives

$$H_{p-1}^{(2)} \ = \ H_m^{(2)} + \sum_{j=1}^{m} (p-j)^{-2} \ \equiv \ 2H_m^{(2)} + 2pH_m^{(3)} \pmod{p^2}.$$

Combining the two displays ($p$ is odd, so $2$ is invertible):

$$-\frac{p}{2} H_{p-1}^{(2)} \ \equiv \ -pH_m^{(2)} - p^2 H_m^{(3)} \ \equiv \ H_{p-1} \pmod{p^3}.$$

Step 2 (the classical congruence, quoted from Glaisher). Glaisher [2] proved

$$H_{p-1}^{(2)} \ \equiv \ \frac{2p}{3} B_{p-3} \pmod{p^2} \qquad (p \geq 5)$$

(his method: $j^{-2} \equiv j^{p-3} - p\, q_p(j)\, j^{p-3} \pmod{p^2}$, where $q_p(j) = (j^{p-1} - 1)/p$ is the Fermat quotient; the power sum $\sum_j j^{p-3} \equiv pB_{p-3} \pmod{p^2}$ comes from the Bernoulli-polynomial summation formula together with the von Staudt–Clausen theorem, and the Fermat-quotient terms are collected by Lerch-type identities). Combining the two steps: $H_{p-1} \equiv -\frac{p}{2} \cdot \frac{2p}{3} B_{p-3} = -\frac{p^2}{3} B_{p-3} \pmod{p^3}$. Final clause: $r_p = 0 \iff u_p \equiv 0 \pmod{p} \iff v_p(H_{p-1}) \geq 3$. ∎

**Remark 7.2.1 (precision of the formulation).** The $u_p$-normalization of Theorem 7.2 is the only meaningful one: by Theorem 3.2(ii), $H_{p-1}$ already contains the factor $p^2$, so "$H_{p-1} \bmod p^2$" vanishes identically and carries no information; only the residue after division by $p^2$ is a signal. All harbinger statistics of this paper (Table B.3) are computed in this normalization; the congruence has been verified for all 428 primes with $5 \leq p \leq 2999$, with zero mismatches.

### 7.3 Erasure and re-infiltration

Lemma 2.3 gave the bound $v_q(\mathrm{den}(H_r)) \le \lfloor \log_q r \rfloor$ together with the equality condition. The exact row-by-row recursion is as follows.

**Proposition 7.4 (erasure criterion).** Write $H_{r-1} = A/B$ in lowest terms ($A, B > 0$), so $H_r = (rA + B)/(rB)$. Then for every prime $q$,

$$v_q\big(\mathrm{den}(H_r)\big) \;=\; \max\big(0,\; v_q(rB) - v_q(rA+B)\big).$$

Put $a := v_q(B)$. Erasure ($v_q(\mathrm{den}(H_r)) < a$) occurs if and only if $v_q(rA+B) > v_q(r)$; a necessary condition is

$$v_q(r) = a \qquad \text{and} \qquad rA \;\equiv\; -B \pmod{q^{a+1}}.$$

**Proof.** Let $g = \gcd(rA+B,\, rB)$, so $\mathrm{den}(H_r) = rB/g$. For any $q$: $v_q(g) = \min(v_q(rA+B),\, v_q(rB))$ (note that the coprimality of $A$ and $B$ prevents $q$ from dividing both: if $v_q(B) > 0$ then $v_q(A) = 0$, whence $v_q(rA) = v_q(r)$; if $v_q(B) = 0$ the same formula holds). Hence $v_q(\mathrm{den}(H_r)) = v_q(rB) - \min(v_q(rA+B), v_q(rB)) = \max(0,\; v_q(rB) - v_q(rA+B))$. Erasure is equivalent to $v_q(rB) - v_q(rA+B) < a$, i.e., $v_q(rA+B) > v_q(r)$. If $v_q(rA) \ne v_q(B)$, then $v_q(rA+B) = \min(v_q(rA), v_q(B)) = \min(v_q(r), a) \le v_q(r)$ and no erasure is possible; hence necessarily $v_q(r) = a$ (and then $v_q(B) = a > 0$ forces $v_q(A) = 0$). Under $v_q(r) = a$, write $r = q^a r'$, $B = q^a B'$ with $q \nmid r'B'$; then $rA + B = q^a(r'A + B')$, and erasure $\iff q \mid r'A + B'$, i.e., $rA \equiv -B \pmod{q^{a+1}}$ after dividing both sides by $q^a$. ∎

**Remark 7.4.1 (rarity and examples).** The necessary condition $v_q(r) = v_q(\mathrm{den}(H_{r-1}))$ is already stringent: it requires the new row index $r$ to carry exactly the same $q$-power as the current denominator. An exact scan (Algorithm A.4; $r \le 2000$, all primes $q$) records exactly 40 complete erasures (plus 15 partial ones, see Table B.4); the first five complete erasures are

$$(r, q) = (6, 3),\; (20, 5),\; (21, 3),\; (33, 11),\; (42, 7).$$

Worked example: $\mathrm{den}(H_5) = 60 = 2^2 \cdot 3 \cdot 5$ (contains 3), $\mathrm{den}(H_6) = 20 = 2^2 \cdot 5$ (3 erased: $rA + B = 6 \cdot 137 + 60 = 882 = 2 \cdot 3^2 \cdot 7^2$, $v_3(882) = 2 > v_3(6) = 1$ ✓); then $\mathrm{den}(H_7) = 420$ contains 3 again — re-infiltration. Re-infiltration does not contradict Theorem 7.1: the first appearance of 3 remains at $r = 3$.

## §8 The double ledger: denominator account and numerator account

The division of labor revealed by Theorems 3.1 and 3.2 — the denominator marks positions, the numerator carries congruences — connects, on the macroscopic scale, to two classical objects of analytic number theory.

### 8.1 The denominator ledger: $\psi(x)$, the prime number theorem, and the Riemann hypothesis

By Lemma 2.3, $\ln \mathrm{den}(H_r) = \sum_q \max(0,\; \lfloor \log_q r \rfloor - \varepsilon_{r,q}) \ln q$, where $\varepsilon_{r,q} \ge 0$ is the accumulated erasure. Comparing with

$$\psi(r) = \ln L_r = \sum_q \lfloor \log_q r \rfloor \ln q,$$

one obtains the identity

$$\psi(r) - \ln \operatorname{den}(H_r) \;=\; \sum_q \varepsilon_{r,q} \ln q \;\geq\; 0,$$

i.e., "denominator ledger = Chebyshev ledger − erasure ledger". The rarity of erasures (Remark 7.4.1: only 40 complete ones for $r \leq 2000$) means that the two ledgers asymptotically coincide. The growth law of the harmonic denominators therefore rigorously inherits two milestones of analytic number theory:

**Proposition 8.1 (asymptotic status of the denominator ledger).** (i) (in the tradition of Chebyshev [3]; standard proof in [13]) the prime number theorem is equivalent to $\psi(x) \sim x$; (ii) (von Koch [4]) the Riemann hypothesis is equivalent to

$$\psi(x) - x \;=\; O\big(\sqrt{x}\,\ln^2 x\big).$$

Consequently $\ln \operatorname{den}(H_{\lfloor x \rfloor}) = x + o(x)$ holds unconditionally (the erasure ledger being $o(x)$ is supported by the $40/2000$ density extrapolation, but a proof of its $o(x)$ order requires control of the erasure distribution; see §11, Problem 2).

**Remark 8.1.1.** The logarithmic factor in the von Koch equivalence is $\ln^2 x$, not $\ln x$: the stronger $O(\sqrt{x}\ln x)$ fails (oscillation results of the shape $\psi(x) - x = \Omega_\pm(\sqrt{x}\ln\ln\ln x)$ are known; see the relevant chapters of [13]). Adopt the standard form $O(\sqrt{x}\ln^2 x)$, which is strictly equivalent to RH.

### 8.2 The numerator ledger: empirical distribution of the harbinger residues

The residue $r_p \in \{0, \dots, p-1\}$ of Definition 7.2 folds the Wolstenholme-type congruence into a ledger with one cell per prime. $r_p = 0$ characterizes the Wolstenholme primes; below $10^9$ there are only two [7]. For all 428 primes with $5 \leq p \leq 2999$, the empirical statistics of $r_p/p$ are: sample mean $0.4826$ (uniform expectation $1/2$); chi-square statistic over 27 equal bins of $[0,1)$ equal to $28.98$ (26 degrees of freedom, $p$-value $\approx 0.312$); zero empty bins (of 27; expectation $15.85$ per bin). The data are in Table B.3. The following conjecture is natural, but it remains unproved:

**Conjecture 8.2 (equidistribution of harbinger residues).** The sequence $\{r_p/p\}_{p\geq 5}$ is equidistributed in $[0,1]$ with respect to natural density; in particular, the counting function of $r_p = 0$ (Wolstenholme primes) satisfies $\#\{p \leq x : r_p = 0\} \sim \ln\ln x$ in order of magnitude (compatible with only two instances below $10^9$).

## §9 The no-closed-form principle

The ledgers of the preceding sections are additive; this section explains why such additive ledgers cannot be sealed into any elementary closed form. Two propositions rule out the two shortcuts of "logarithmic linear combinations" and "Mills-type constants" respectively.

**Proposition 9.1 ($\mathbb{Q}$-linear independence of the prime logarithms).** The set $\{\ln p :\ p \text{ prime}\}$ is linearly independent over $\mathbb{Q}$.

**Proof.** Suppose a finite rational combination $\sum_p a_p \ln p = 0$. Clearing denominators gives an integral relation $\sum_p b_p \ln p = 0$ ($b_p \in \mathbb{Z}$, finite support); exponentiating, $\prod_p p^{b_p} = 1$. Separating positive and negative exponents, $\prod_{b_p>0} p^{b_p} = \prod_{b_p<0} p^{-b_p}$, both sides being positive integers; the fundamental theorem of arithmetic (unique factorization) forces every exponent to vanish, i.e., all $b_p = 0$. ∎

**Remark 9.1.1.** Proposition 9.1 is the logarithmic translation of the fundamental theorem of arithmetic, and its proof is trivial, but the corollary is not: any attempt to write $\{H_p\}$ or $\{\mathrm{den}(H_p)\}$ in "elementary closed form in finitely many basic constants" must first compress the $\mathbb{Q}$-independent set $\{\ln p\}$ — a task of the same difficulty as unique factorization itself. This is the first ground for the no-closed-form position below.

**Proposition 9.2 (Mills-type tautology).** There exists a real constant $A > 1$ such that $\lfloor A^{3^n} \rfloor$ is prime for every $n \geq 1$ (Mills [9]); but the decimal expansion of any such $A$ encodes, in a computable way, exactly the primes it outputs, so such a "formula" does not constitute a compressed representation of the prime sequence.

**Proof.** Existence is Mills' theorem (an iterative construction based on prime-gap estimates of Ingham's type $p_{n+1} - p_n \ll p_n^{5/8+\varepsilon}$). For the tautology: in Mills' construction $A = \lim_n P_n^{3^{-n}}$, where $P_n$ is the $n$-th prime satisfying the inequality constraints; given $A$ and $n$, reading off $P_n = \lfloor A^{3^n} \rfloor$ merely undoes the construction. The information-theoretic check: if $A$ were generated by a short program and output all the primes, the algorithmic information of the prime sequence would be compressed to $O(1)$ plus the input $n$ — this does not directly contradict the $\mathbb{Q}$-independence of $\{\ln p\}$, but Mills' construction itself shows that the definition of $A$ recursively calls the prime sequence, so no compression is achieved. ∎

**Remark 9.2.1.** The position of this paper is therefore: the transfer-triangle system does not "compute" the primes; it transcribes their multiplicative structure into valuation events of the denominator (Theorem 7.1) and a congruence ledger of the numerator (Theorem 7.2). The transcription is faithful (it is an equivalence), but it is not a compression — §10 makes the computational boundary of the system precise.

## §10 The boundary of computational power

### 10.1 Non-regularity of the bare syntax

Encode whether a new-letter event occurs at row $r$ as a $0/1$ sequence: by Theorem 7.1, this is exactly the prime indicator sequence $\chi(r) = 1[r \text{ is prime}]$.

**Proposition 10.1 (the bare syntax generates no regular language).** (i) The sequence $\chi$ is not regular (its set of 1-positions is accepted by no finite automaton); (ii) more generally, the state of the pure D/B recursion (Definition 5.1) is completely determined by the scalar parameter $r$, and its orbit is a one-parameter curve: under any finite partition of the state space, the resulting language is the image of a single orbit and cannot recognize languages of the type $\{a^n b^n\}$.

**Proof.** (i) Suppose $\chi$ were regular. By the pumping lemma, the gaps between consecutive 1-positions in an infinite regular $0/1$ sequence are bounded (by the number of states of the automaton). But for every $n$, the consecutive composites $n! + 2,\ n! + 3,\ \ldots,\ n! + n$ form a prime-free segment of length $n - 1$, so prime gaps are unbounded — contradiction. (ii) The D/B recursion has no external input: each step of $P^{(r+1)} = B_r(P^{(r)})$ is determined by the function $\lambda_r$ of $r$, and the state space is effectively one-dimensional (the parameter $r$). Under any finite partition, the symbolic sequence is a single fixed infinite word, whose finite-subword complexity grows polynomially in $n$ (the visitation pattern of the curve $\{z^{(r)}\}$ through the partition cells is controlled by the monotone quantity $1/\sqrt{H_r}$); it cannot realize bracket-matching languages, which require unbounded counting. ∎

### 10.2 Turing completeness of the minimal extension

The converse of Proposition 10.1 is: adjoining one data-dependent primitive — conditional duplication of a unit of storage — gives Turing completeness, with no division, no external tape, no external clock (Appendix C).

**Proposition 10.2 (minimal extension = Turing completeness).** Adjoin to the D/B recursion a finite control graph with two instruction forms on a prime-indexed register vector $x \in \mathbb{N}^{(P)}$: unit addition $x \mapsto x + e_b$; conditional duplication $C_{a \to b}$: if $x_a \geq 1$ move one unit from $a$ to $b$, else branch. Fresh coordinates are supplied deterministically by the variable frame (Lemma C.2); the resulting system is Turing complete.

**Proof (sketch; full details in Appendix C).** A two-counter Minsky machine [11] configuration $(l; c_1, c_2)$ is encoded as $(l, c_1 e_{q_1} + c_2 e_{q_2}, r)$; $\mathrm{INC}_i$ becomes unit addition at $q_i$, $\mathrm{DEC}_i$/zero-test becomes $C_{q_i \to w}$ with its two continuations; induction on the step count identifies the two runs step by step, uniformly in the machine (Theorem C.3). Under the naming $N(x) = \prod_p p^{x_p}$, a conditional duplication is $N \mapsto N{\cdot}p_b/p_a$, integral iff $p_a \mid N$ — so division never occurs as an operation, and every guarded Fractran multiplication [10, 17] compiles into finitely many guarded transfers with restore-on-failure (Lemma C.1). The guards are decidable zero-tests and the updates computable rational translations, so configurations are uniformly computable in $(n, t)$; the halting set is r.e.-complete, of degree exactly $0'$ (Theorem C.4). ∎

**Remark 10.2.1.** The gap between Propositions 10.1 and 10.2 is one bit of control: a single guarded fork. The guarded unit transfer is the decrement/zero-test atom of Minsky [11] and Lambek [15]; Melzak's ternary XYZ [16] is its block-transfer ancestor. Disjoint-coordinate instructions commute: an asynchronous, resynchronization-free parallel reading (Lemma C.6); the system is exactly Turing complete, never hyper-Turing (Theorem C.4).

## §11 Open problems

1. **Equidistribution of the harbinger residues.** Prove or disprove Conjecture 8.2: $\{r_p/p\}$ is equidistributed in $[0, 1]$. This is equivalent to controlling the distribution of $B_{p-3} \bmod p$ in Glaisher's congruence (Theorem 7.2), a problem of the same kin as the $p$-adic distribution of Bernoulli numerators; the available methods (Kummer congruences, $p$-adic $L$-functions) seem insufficient for a direct attack.
2. **Distribution of erasures.** The empirical density of the 40 complete erasures ($r \leq 2000$) is about $2\%$, concentrated in the window $v_q(r) = v_q(\mathrm{den}(H_{r-1}))$ (Proposition 7.4). Questions: does the counting function $E(x)$ of erasure events satisfy $E(x) = o(x)$? Is $E(x) \asymp \ln x$? This control is exactly the gap in promoting the denominator-ledger asymptotics of Proposition 8.1 to an unconditional theorem.
3. **Wolstenholme primes.** Does $r_p = 0$ have infinitely many solutions? Only $16843$ and $2124679$ are known below $10^9$ [7]; under Conjecture 8.2 the expected count is $\sim \ln\ln x$, compatible with observation.
4. **The composite-numerator problem.** Prove or disprove Conjecture 3.4 ($n$ composite $\Rightarrow n \nmid \mathrm{num}(H_{n-1})$; verified here up to $n \leq 2000$). It is adjacent to Eswarathasan and Levine's conjectures [6] on $\{n : H_n \in \mathbb{Z}_p\}$.

5. **Explicit simulation bounds.** The Fractran-style simulation of Proposition 10.2 uses arbitrarily large intermediate integers. Question: if the total bit-length of the state is restricted to $O(\mathrm{poly}(r))$, can the system simulate universal computation with polynomial overhead? This amounts to asking whether prime bookkeeping natively carries a class of "arithmetic-circuit lower bounds".

## Appendix A Algorithms

The following four algorithms constitute the reproducible basis of every numerical assertion in the paper. Conventions: rationals are stored as arbitrary-precision reduced fractions; modular inverses use the extended Euclidean algorithm; complexity is counted in arithmetic operations. The reference implementation (Python, <100 lines) runs in a few seconds at the scales $r \le 2000$ and $p \le 2999$.

**Algorithm A.1 (exact harmonic state generation).**

```
Input: r ≥ 1
Output: numerator and denominator of H_r (reduced), state vector P^(r)
H ← 0/1
for k = 1..r:
    H ← H + 1/k            # automatic reduction
P ← [ (1/k)/H for k=1..r ]  # exact fraction vector
return H.num, H.den, P
```

**Algorithm A.2 (denominator anchoring and the composite check; Theorem 3.1, Proposition 3.3).**

```
Input: bound N
H ← 0/1
for r = 1..N:
    H ← H + 1/r
    if r is prime:
        assert: v_r(H.den) = 1 and r ∤ H.num          # Theorem 3.1(i)(ii)(iv)
    else if r ≥ 4:
        assert: r ∤ H.num                             # Proposition 3.3
return: mismatch list (empty for N = 2000 in this paper)
```

**Algorithm A.3 (harbinger residues; the numerical side of Theorem 7.2).**

```
Input: prime p ≥ 5
Output: r_p ∈ {0,...,p-1}, with Glaisher's congruence verified
M ← p³;  s ← 0
for j = 1..p-1:
    s ← (s + j^{-1} mod M) mod M        # modular inverse, O(log M)
assert: p² | s                            # Wolstenholme
u ← (s / p²) mod p                        # harbinger residue r_p
B ← B_{p-3} mod p                         # Akiyama-Tanigawa recursion
assert: u ≡ -B/3 (mod p)                  # Theorem 7.2
return u
```

**Algorithm A.4 (erasure scan; Proposition 7.4).**

```
Input: bound N
H ← 0/1;  D_prev ← 1;  event table E ← ∅
for r = 1..N:
    H ← H + 1/r;  D ← H.den
    for q ∈ primeFactors(D_prev):
        a ← v_q(D_prev);  b ← v_q(D)
```

```
        if b < a:  record (r, q, a, b) into E   # b=0: complete erasure
        assert: v_q(r) = a                        # necessary condition, Prop. 7.4
    D_prev ← D
return: E (for N = 2000 in this paper: 40 complete erasures, 15 partial)
```

## Appendix B  Data tables

All values are independently verified by Algorithms A.1–A.4 in exact rational arithmetic, or in modular arithmetic at the stated modulus; floating-point columns are double-precision displays with the last digit rounded.

**Table B.1  The first nine prime rows: $H_p$, reduced denominator, compression ratio, and stepwise angle**

| $p$ | $H_p$ **(reduced)** | $\mathrm{den}(H_p)$ | $\lambda_p = H_p/H_{p+1}$ | $\theta_p$ **(deg)** |
|---|---|---|---|---|
| 2 | $3/2$ | 2 | 0.8181818182 | 25.239402 |
| 3 | $11/6$ | 6 | 0.8800000000 | 20.267901 |
| 5 | $137/60$ | 60 | 0.9319727891 | 15.118739 |
| 7 | $363/140$ | 140 | 0.9540078844 | 12.383715 |
| 11 | $83711/27720$ | 27720 | 0.9731460922 | 9.431691 |
| 13 | $1145993/360360$ | 360360 | 0.9780325381 | 8.523453 |
| 17 | $42142223/12252240$ | 12252240 | 0.9841047675 | 7.242915 |
| 19 | $275295799/77597520$ | 77597520 | 0.9861023852 | 6.770237 |
| 23 | $444316699/118982864$ | 118982864 | 0.9889652733 | 6.029835 |

**Table B.2  Compression ratio, stepwise angle, and total budget (Theorems 6.2, 6.3; $\theta_r$ in degrees, $\Theta_r$ and the gap in radians)**

| $r$ | $\lambda_r$ | $\theta_r$ | $\Theta_r = \arccos(1/\sqrt{H_r})$ | $\pi/2 - \Theta_r$ |
|---|---|---|---|---|
| 1 | 0.6666666667 | 35.264390 | 0.0000000000 | 1.5708 |
| 2 | 0.8181818182 | 25.239402 | 0.6154797087 | 0.9553 |
| 3 | 0.8800000000 | 20.267901 | 0.7398807744 | 0.8309 |
| 5 | 0.9319727891 | 15.118739 | 0.8476022948 | 0.7232 |
| 10 | 0.9698964294 | 9.991601 | 0.9467678545 | 0.6240 |
| 20 | 0.9869370749 | 6.562857 | 1.0154803606 | 0.5553 |
| 50 | 0.9956608423 | 3.776939 | 1.0798664456 | 0.4909 |
| 100 | 0.9980949664 | 2.501566 | 1.1162416838 | 0.4546 |
| 200 | 0.9991543228 | 1.666427 | 1.1456411857 | 0.4252 |
| 400 | 0.9996205714 | 1.116131 | 1.1700135101 | 0.4008 |

**Table B.3  Harbinger-residue statistics (the 428 primes $5 \le p \le 2999$; $r_p$ computed by Algorithm A.3, Glaisher's congruence verified pointwise with zero mismatches)**

| Statistic | Value | Remark |
|---|---:|---|
| Number of primes | 428 | $p \in [5, 2999]$ |
| Count of $r_p = 0$ (Wolstenholme) | 0 | consistent with [7] (only two below $10^9$) |
| Sample mean of $r_p/p$ | 0.4826 | uniform expectation 0.5 |
| Sample s.d. of $r_p/p$ | 0.2875 | uniform expectation $1/\sqrt{12} \approx 0.2887$ |
| Chi-square (27 equal bins) | 28.98 | 26 d.f., $p$-value $\approx 0.312$ |
| Empty bins (of 27) | 0 | expectation 15.85 per bin |

**Table B.4 Erasure events ($r \leq 2000$, Algorithm A.4): exactly 40 complete erasures and 15 partial ones; the necessary condition $v_q(r) = v_q(\mathrm{den}(H_{r-1}))$ holds for all 55 events**

| Type | Count | First five $(r, q, a \to b)$ |
|:---:|:---:|---|
| Complete ($b = 0$) | 40 | $(6, 3, 1{\to}0)$, $(20, 5, 1{\to}0)$, $(21, 3, 1{\to}0)$, $(33, 11, 1{\to}0)$, $(42, 7, 1{\to}0)$ |
| Partial ($b \geq 1$) | 15 | $(18, 3, 2{\to}1)$, $(54, 3, 3{\to}2)$, $(63, 3, 2{\to}1)$, etc. |

**Table B.5 Denominator ledger versus Chebyshev ledger (Proposition 8.1; the difference is the erasure ledger $\sum_q \varepsilon_{r,q} \ln q$)**

| $r$ | $\psi(r) = \ln \mathrm{lcm}(1..r)$ | $\ln \mathrm{den}(H_r)$ | Erasure ledger |
|---:|---:|---:|---:|
| 10 | 7.8320 | 7.8320 | 0 |
| 50 | 49.4854 | 49.4854 | 0 |
| 100 | 94.0453 | 90.8264 | 3.2189($= \ln 25$, event $(100, 5, 2{\to}0)$) |
| 500 | 501.6521 | 497.3346 | 4.3175 |
| 1000 | 996.6809 | 996.6809 | 0 |
| 2000 | 1994.4516 | 1993.3530 | 1.0986($= \ln 3$) |

Note: the value $\mathrm{lcm}(1, \ldots, 23) = 5354228880$ used in Theorem 6.4 was also verified directly.

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# Appendix C Full proofs for §10.2

This appendix gives the complete definitions and proofs behind Proposition 10.2. Throughout, $\mathbb{N}^{(P)}$ denotes the abelian group of finitely supported vectors on prime-indexed coordinates, $e_a$ its standard basis vectors, and $D/B$ the recursion of Definition 5.1 with $\lambda_r = H_r/H_{r+1}$.

## C.1 The extended system

The extended system has three components. (i) *Bookkeeping.* The bare $D/B$ recursion runs autonomously, one row per tick; its $r$-th row re-bases the ambient frame to $\Delta^{r-1}$. (ii) *Registers.* At tick $t$ (bookkeeping row $r_t$) the register vector $x \in \mathbb{N}^{(P(r_t))}$ has finite support on the prime-indexed coordinates supplied so far, $P(r) = \{p \text{ prime} : p \le r\}$; Lemma C.2 shows that the frame supplies coordinate $p$ exactly at row $r = p$. (iii) *Control.* A finite directed graph whose nodes are: HALT; *unit addition* $x \mapsto x + e_b$ with a single continuation; and *conditional duplication* $C_{a\to b}$: if $x_a \ge 1$ then $x \mapsto x - e_a + e_b$ and continue $l_0$, else $x$ is unchanged and continue $l_1$.

A configuration is $(l, x, r)$. One global step executes one register instruction and advances one bookkeeping row. An instruction addressing coordinate $p$ is legal only once $p \le r_t$; every finite program mentions finitely many coordinates and $r_t \to \infty$, so every needed coordinate is eventually present (the clock initialization of Theorem C.3 absorbs the startup delay, so the support constraint holds from the first genuine step on). No external storage is postulated — the frame itself generates, deterministically, an unlimited supply of fresh, pairwise linearly independent register coordinates — and no external clock either: the row index $r_t$ is advanced by the system's own bookkeeping recursion, is never queried by the control, and for any fixed program is dispensable beyond the compile-time constant $r_0 = \max Q$ of Theorem C.3. *Input:* $n \mapsto n \cdot e_{q_{\text{in}}}$ for a designated input coordinate $q_{\text{in}}$; *output:* $x_{q_{\text{out}}}$ when HALT is reached; runs that never reach HALT yield no output. Input and output are unary; $k$-ary functions are handled by pairing, so the unary convention suffices. Under the naming map $N(x) = \prod_p p^{x_p} \in \mathbb{N}_{\ge 1}$ (a bijection by unique factorization), unit addition is $N \mapsto p_b \cdot N$ and conditional duplication is $N \mapsto N \cdot p_b/p_a$, integral iff $p_a \mid N$. Division occurs only in this multiplicative naming of the state, never as an operation of the system: operationally a conditional duplication removes one unit at $a$ and adds one unit at $b$.

> **Lemma C.1 (no division; exact Fractran correspondence).** (i) Every conditional rational multiplication "if $n \mid N$ then $N \mapsto N \cdot m/n$, else continue $l_1$" compiles into finitely many conditional duplications and unit additions with the same success/failure continuations. (ii) Conversely, every program of the extended system compiles into a Fractran program with step-for-step identical runs. (iii) The guarded unit transfer is the decrement-with-zero-test atom of Minsky [11] and of Lambek's abacus (1961) [15]; Melzak's ternary $XYZ$ operation (1961) is its block-transfer ancestor [16].

**Proof.** (i) Write $n = \prod_p p^{\beta_p}$ and $m = \prod_p p^{\alpha_p}$. Reserve $s = \sum_p \beta_p$ dedicated waste coordinates $w_1, \dots, w_s$, one for each unit to be removed. *Removal phase*: attempt the $\beta_p$ removals of each prime $p \mid n$ one unit at a time, sending each removed unit to its own dedicated waste coordinate; the invariant is that every $w_j$ is empty at entry to the composite and holds at most one unit during it. If every removal succeeds, *addition phase*: perform the $\sum_p \alpha_p$ unit additions for $m$; then *flush phase*: the transfers $C_{w_j \to g}$ (with $g$ a garbage coordinate, never read) empty each waste — each guard succeeds since $w_j = 1$ just after its removal — restoring the entry invariant, and continue $l_0$. If the guard fails at removal step $k+1$, so $n \nmid N$, *restore phase*: the static reverse chain over $w_k, \dots, w_1$ — determined by the failure point, not guard-driven — returns exactly the units removed so far (each $w_j$, $j \le k$, holds exactly one unit and remembers its origin by construction), restoring $N$ exactly, and continue $l_1$. Because the wastes are dedicated per step and flushed after each success, no units of different origins or of earlier firings are ever confounded. The composite has exactly the all-or-nothing semantics of the guarded rational multiplication.

(ii) Encode control states by distinct primes $s_l$. Conditional duplication $C_{a\to b}$ at node $l$ with continuations $l_0, l_1$ becomes the ordered fraction pair $\frac{p_b s_{l_0}}{p_a s_l}$ (success) and $\frac{s_{l_1}}{s_l}$ (failure); unit addition $x \mapsto x + e_a$ becomes $\frac{p_a s_{l'}}{s_l}$. The state primes are distinct from all data and waste primes, and every fraction's denominator carries its node's state prime while its numerator carries the successor's, so exactly one state prime divides the current integer at any time, with exponent exactly 1; hence fractions of different nodes never compete, and within a node the success fraction precedes the failure fraction, which under Fractran's first-applicable rule [10, 17] fires iff the success guard fails — this is the zero-test. No fraction has $s_{\mathrm{HALT}}$ in its denominator, so Fractran termination coincides with reaching HALT. A routine induction identifies runs step for step.

(iii) For [11, 15, 16] see the bibliography. We keep the name *conditional duplication* because the restore-loop compositions of (i) realize conditional copy, the form in which the atom is actually used. ■

**Lemma C.2 (the frame supplies fresh registers deterministically).** (i) The direction $e_{r+1}$ adjoined by $B_r$ is a fresh standard basis vector, pairwise linearly independent of — indeed orthogonal to — every previously present coordinate direction. (ii) The adjunction is deterministic and effective: $\lambda_r = H_r/H_{r+1}$ is an explicitly computable rational number and $1 - \lambda_r = \frac{1/(r+1)}{H_{r+1}} > 0$, so the new coordinate enters with strictly positive weight. (iii) Let $L_r = \mathrm{lcm}(1, \ldots, r)$. Then $v_p(L_r) = \lfloor \log_p r \rfloor$; in particular a prime $p$ divides $L_r$ iff $p \le r$, first entering exactly at $r = p$, while prime powers $p^k$ only raise the $p$-exponent.

**Proof.** (i) $B_r(x) = D_r(x) \oplus (1 - \lambda_r)e_{r+1} \in \Delta^r$, and $e_{r+1}$ is the $(r+1)$-st standard basis vector of $\mathbb{R}^{r+1}$, pairwise linearly independent of $e_1, \ldots, e_r$. (ii) $H_{r+1} = H_r + \frac{1}{r+1}$ gives the displayed identity, and harmonic numbers are computable rationals. (iii) $v_p(\mathrm{lcm}(1, \ldots, r)) = \max_{1\le k\le r} v_p(k) = \max\{e : p^e \le r\} = \lfloor \log_p r \rfloor$. This exponent is 0 iff $p > r$ and turns 1 exactly at $r = p$; at $r = p^k$ it increments to $k$. ■

**Theorem C.3 (universality, direct).** The extended system computes every unary partial recursive function, and the simulation is uniform in the program.

**Proof.** By Minsky [11], two-counter machines (instructions $\mathrm{INC}_i$, $\mathrm{DEC}_i$/zero-test, $i = 1, 2$; finite state set) compute every partial recursive function. Fix such a machine $M$ with counters $c_1, c_2$ and a halt state. Reserve register coordinates $q_1, q_2$ and a scratch coordinate $w$, and encode the $M$-configuration $(l; c_1, c_2)$ as the extended configuration $(l,\ c_1 e_{q_1} + c_2 e_{q_2},\ r)$. Instructions compile as follows: $\mathrm{INC}_i$ at $l$ becomes unit addition at $q_i$ with the continuation of $l$; $\mathrm{DEC}_i$/zero-test at $l$ with continuations $l_{\mathrm{dec}}, l_{\mathrm{zero}}$ becomes conditional duplication $C_{q_i\to w}$ with success continuation $l_{\mathrm{dec}}$ and failure continuation $l_{\mathrm{zero}}$ (the removed unit is parked in $w$, which is never read). *Clock legality:* let $Q$ be the finite set of coordinates mentioned by the compiled program, together with $q_{\mathrm{in}}$; initialize the bookkeeping clock at $r_0 = \max Q$ (equivalently, prepend $\max Q$ idle ticks via a skip node), a finite compile-time constant whose skipped rows are computable constants. From the first genuine instruction on, every addressed coordinate — including the input coordinate — is present in the frame (Lemma C.2(iii)). *Correctness:* by induction on $t$, after $t$ compiled steps the extended configuration is exactly the image of the $M$-configuration after $t$ steps: unit addition adds $e_{q_i}$; conditional duplication subtracts $e_{q_i}$ iff $x_{q_i} \ge 1$, i.e. iff $c_i \ge 1$, and branches on $x_{q_i} = 0$, i.e. iff $c_i = 0$. Hence the extended run reaches HALT iff $M$ halts, and then $x_{q_1} = c_1$ is the output. The compilation is effective and uniform in $M$. (It inherits Minsky's exponential slowdown — a matter of efficiency, not of power.) *Control in the registers:* control may equivalently be carried by state coordinates $s_l$ held invariantly at unit mass; each instruction then compiles to at most three legal nodes — the data operation, and one unconditional leg $C_{s_l\to s_{l'}}$ per continuation, whose guard holds invariantly because exactly one state register equals 1 — and the zero-test ordering of Lemma C.2(ii) recovers Fractran's first-applicable semantics [10]. Both presentations compute the same transition function. ■

**Theorem C.4 (exactness — nothing beyond $0\prime$.** The halting set of the extended system is many-one complete for the recursively enumerable sets: the system is exactly Turing complete. The same holds for any finite enrichment by instruction forms whose guards are decidable predicates of the current configuration and whose updates are computable rational affine maps.

**Proof.** *Computability of runs.* The configuration $(l_t, x_t, r_t)$ is uniformly computable from $(n, t)$: induction on $t$; the guard at $l_t$ is a zero-test on the current register vector, decidable; the update is a translation of $x_t$ by $\pm$ unit vectors, computable; the bookkeeping row is a computable rational sequence in $r$ (harmonic numbers, induction on $r$. *Semi-decidability:* acceptance is semi-decidable by running the computation. *Completeness:* the compilation of Theorem C.3 is effective and uniform in $M$, so the halting set is r.e.-complete, of degree exactly $0\prime$. *Enrichments:* the same induction goes through verbatim for any finite table of instruction forms with guards decidable in the current configuration and computable rational affine updates; the enriched halting set remains r.e., hence $\leq 0\prime$, and universality gives equality. *What would break the ceiling:* an infinite rule table (the induction then lacks a finite description), a guard not decidable in the current configuration (e.g. a limit predicate on the orbit), or a non-computable constant in an update law. The limit questions of §8–§9 (e.g. Conjecture 8.2) are external analytic statements about the orbit, not instructions of the machine; their possible independence is the ordinary Gödelian phenomenon, not hypercomputation. ■

**Lemma C.6 (commutation; asynchrony).** Let $\sigma, \sigma\prime$ be two instruction instances that do not interfere: neither instance *reads* a coordinate the other *writes* (for $C_{a\to b}$ the read set is $\{a\}$ and the write set $\{a, b\}$; unit addition at $b$ reads nothing and writes $\{b\}$. If both guards hold at the register vector $x$, then firing $\sigma$ then $\sigma\prime$, or $\sigma\prime$ then $\sigma$, yields the same register vector, and each guard still holds after the other instance fires.

**Proof.** Each instance is a guarded translation $x \mapsto x + \delta$ with $\delta \in \{+e_b,\ -e_a + e_b\}$, its guard, where present, a zero-test on a single coordinate. Non-interference says precisely that neither guard reads a coordinate the other writes, so both guards are invariant under the other's firing; translations commute unconditionally, and the non-negativity of every intermediate value is secured by the guard invariance. ■

Corollary C.7 (synchronization-free interleaving). Any finite family of enabled, pairwise independent instruction instances may be fired in any order — or concurrently — with the same final register vector: the causal order of a computation, over the instruction instances within independent phases or across parallel components, is a partial order (a Mazurkiewicz trace whose independence relation is the non-interference of Lemma C.6), not a clocked total order; within one component the control sequence is genuinely total, as it must be. This has two uses. (i) Inside the composites of Lemma C.1(i), the flush transfers $C_{w_j\to g}$ for different $j$ are pairwise independent — each reads only its own $w_j$, and the sink $g$ is written by all but read by none — and independent phases generally may execute in any order. (ii) A finite parallel composition of extended systems on coordinate-disjoint register lattices needs no shared clock and no handshake: each component keeps its own local computation order, dependence arises exactly at shared coordinates, and under any fair schedule (equivalently: on jointly terminating runs) the joint input–output map is the tuple of the component maps. Since the sequential run of Theorem C.3 is a distinguished — and fair — linearization of this partial order, universality is unaffected; since every finite interleaving is effectively simulable, the ceiling of Theorem C.4 is likewise unaffected.

**Proposition C.5 (structural portrait).** (i) *Mass conservation (an $L^1$-analogue of unitarity).* Conditional duplication conserves the total mass $\|x\|_1$, and unit addition is the unique mass source; the parked-unit coordinates act as ancilla registers. This is a classical affine $L^1$-analogue of unitarity, not Hilbert-space unitarity. (ii) *Functional completeness.* By Theorems C.3–C.4 the system computes exactly the partial recursive functions — completeness in the recursion-theoretic sense, not in Post's Boolean-clone sense. (iii) *Variable frame.* The carrier re-bases each row ($\Delta^{r-1}$) and fresh prime coordinates enter deterministically (Lemma C.2), unlike fixed-finite-dimensional models; yet finitely many coordinates suffice for universality (Theorem C.3 uses three), so the growing frame is architectural — self-supplied storage — while the computational power comes exactly from the guarded control flow. This is the precise content of the Proposition 10.1/10.2 boundary. (iv) *Asynchrony and parallelism.* On disjoint coordinates the instructions commute (Lemma C.6): each branch of a parallel composition carries its own local computation order, asynchronous and never resynchronized, and any fair schedule refining the causal partial order computes the same map — so both the universality (Theorem C.3) and the exactness (Theorem C.4) survive the asynchronous reading unchanged.

**Proof of the minimality gloss.** Without any guarded fork, every instruction is an unconditional translation, so the control path — and hence the computed map — is independent of the input: on unary input $n$ the system computes $n \mapsto n + c$ for a fixed $c$ (or a constant, if $q_{\mathrm{in}} \neq q_{\mathrm{out}}$, or diverges. No universality can arise. Thus some query primitive is necessary, and a single one-bit guarded fork suffices by Theorem C.3; this is the sense in which the extension is minimal. ■